

Teacher Reflection Lab 3.0: A Scenario-Based Platform for Teacher Self-Awareness and Classroom Decision-Making

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Project Pitch

For our final project, we plan to continue developing our second mini-project, Teacher Reflection Lab 2.0. Teacher Reflection Lab is an online practice platform designed for teachers, especially novice and pre-service teachers, to reflect on their teaching identity and practice classroom decision-making in a low-risk environment. The main learning goal is to help teachers build self-awareness, connect their teaching tendencies to real classroom choices, and practice responding to complex classroom situations before they encounter them with real students.

Our current version already introduces the idea of teacher personality, scenario simulation, and reflective feedback. For the final version, we want to make the platform feel more complete and coherent by connecting three parts into one learning journey: personality test → scenario-based simulation → feedback and growth profile. Instead of treating the personality result as a separate quiz outcome, we want it to become the starting point for personalized practice. Teachers will first receive one of six teaching identity types, then enter classroom scenarios that are especially relevant to that identity, and finally see their choices added back into a growth map or teacher profile.

The six teacher identities we are currently using are Community Energizer, Structured Guide, Creative Explorer, Steady Supporter, Reflective Designer, and Quiet Specialist. These identities are based on Big Five trait combinations, but they are not meant to diagnose teachers or label their whole personality. Instead, they are designed as reflective lenses that help teachers recognize natural strengths and growth edges. This matters because research on teacher personality suggests that teacher dispositions are connected to teaching quality, enthusiasm, effectiveness, and burnout, though personality should be understood in a nuanced and practice-based way rather than as a fixed label (Burić et al., 2023; Kim et al., 2019). For example, teachers' conscientiousness, emotional stability, extraversion, openness, and agreeableness may shape how they organize instruction, build relationships, respond to classroom stress, and create learning environments.

Planned Changes for the Final Product

The first major change is to build the final prototype around the six teaching identities. Each identity will have a clear result page with a short overview, natural strengths, and growth edges. We want the result page to feel supportive and reflective rather than clinical. For example, a Structured Guide may be described as someone who creates safety through clarity and routines, while their growth edge may involve staying flexible when unexpected classroom moments appear. A Steady Supporter may be strong at emotional safety and de-escalation, while their growth edge may involve setting boundaries and making firm decisions. This structure helps teachers see both what they already bring to the classroom and what they can continue practicing.

The second major change is to create six classroom scenarios, with each scenario connected to one teaching identity. These scenarios will not simply ask teachers to choose “right” or “wrong” answers. Instead, they will present realistic classroom tensions where different teacher moves have different consequences. For example, our Sean’s Number simulation asks teachers to respond to a student misconception about whether the number 6 can be both odd and even. The strongest path is not just correcting the student quickly, but making the student’s reasoning visible, testing it with examples, and helping the class build a stronger mathematical rule. This kind of design supports classroom decision-making because teachers must consider accuracy, student thinking, peer discussion, and facilitation at the same time.

The third major change is to add a growth map or teacher profile update. After teachers complete simulations, their choices and feedback will be added to their profiles. This profile will show both their initial teaching identity result and their growth through practice. For example, a teacher might begin as a Creative Explorer and then see that their scenario feedback suggests a need for clearer structure. Another teacher might begin as a Quiet Specialist and see that they are growing in making their thinking more visible to students. This growth map makes the learning process cumulative instead of one-time. It also supports reflection because teachers can track patterns across multiple simulations.

Why We Are Designing It This Way

The “why” behind Teacher Reflection Lab 3.0 stems from the gap between knowing about teaching and actually making classroom decisions. Novice teachers often face situations that are fast, emotional, or uncertain. Research on preservice teacher preparation shows that classroom management and student behavior can be major sources of stress, burnout, and turnover. Simulation-based practice can help by allowing teachers to pause, reflect, receive feedback, and try again without harming real students (Larson et al., 2019). This is exactly the kind of learning environment we want to create: one where teachers can safely experiment with classroom moves before they need to perform them in real time.

We also want to avoid making the platform only about “personality.” A personality result alone can be interesting, but it does not automatically help someone teach better. The learning happens when teachers connect self-awareness to action. This is why each identity must lead to scenario practice. The platform should help teachers ask: What kind of teacher am I becoming? What choices do I tend to make under pressure? What strengths do I rely on? What classroom situations challenge me? What can I try next time?

The Big Five framework gives us a research-based starting point, but our design translates it into teacher-friendly language. Burić et al. (2023) argue that teacher personality should be examined not only across broad domains but also through more specific facets, as these may offer a more detailed understanding of teaching practice. Kim et al. (2019) also found that several Big Five domains are associated with teacher effectiveness and burnout. For our project, this supports the idea that teacher identity can be a meaningful entry point for reflection. However, we will be careful not to suggest that a teacher’s identity determines their effectiveness. Instead, we frame personality as one part of a larger learning process that includes context, practice, feedback, and growth.

Implementation Plan and Design Tasks

To bring the final product to a more finalized state, we need to complete several design tasks. First, we need to revise the personality result pages so that all six identities have consistent formatting, tone, and visual design. Each page should include an identity overview, natural strengths, growth edges, and a clear button that leads into the simulation.

Second, we need to finalize the six classroom scenarios. We will try to base the scenarios on real or open-source classroom examples. If we cannot find enough useful examples, we may use AI-generated drafts, but we will revise them carefully to make them realistic, grounded in teacher challenges, and connected to our learning goals. Each scenario should include a classroom context, a central teaching tension, three meaningful decision points, and a feedback outcome.

Third, we need to connect scenario results back to the teacher profile. This means designing what data to save after each simulation, how feedback is summarized, and how growth is visually represented. The profile should not be too complicated. It should clearly show the teacher's identity, completed scenarios, repeated strengths, and suggested next practice areas.

Fourth, we need to improve the user flow. The platform should guide users from test to results to scenarios to feedback without requiring them to manually browse too much. We will add filters to the scenario selection page so teachers can find scenarios by identity type, classroom challenge, subject area, or difficulty level. We also need to test whether users understand what to do at each step.

Stakeholder Interview Plan

For the stakeholder interview, we would like to interview a professor, teaching fellow, or teacher educator who works with novice teachers or pre-service teachers. This stakeholder is valuable because they understand what beginning teachers often struggle with, what kinds of practice are useful, and what kinds of feedback feel realistic rather than generic. Ideally, this person has experience teaching education courses, supervising student teachers, or supporting classroom-based professional development.

Our main interview questions would include:

1. What classroom situations do novice or pre-service teachers most need to practice before entering the classroom?
2. Do the six teacher identities feel useful, supportive, and appropriate for teacher reflection?
3. Are there any risks in using personality-based language with teachers?
4. What kinds of scenario feedback would feel helpful rather than judgmental?
5. Would a growth map or teacher profile help teachers see their development over time?
6. What parts of the platform might feel confusing, stressful, or unnecessary?
7. How could we make the tool more accessible or adaptable for different teaching contexts?

The feedback we are looking for is both conceptual and practical. Conceptually, we want to know whether the personality-to-simulation structure makes sense as a learning experience. Practically, we want feedback on whether the scenarios feel realistic, whether the report language is understandable, and whether the platform could be used in a teacher education course or professional development setting.

Accessibility, Localization, and Feedback Gathering

Accessibility and localization are important because teaching contexts vary widely. A classroom scenario that feels realistic in one school may not fit another school, grade level, subject area, or cultural context. To address this, we will keep the platform adaptable. Scenario filters will allow teachers to select more relevant cases, and scenario language will avoid assuming a single “normal” classroom environment. We will also keep the design keyboard- and mouse-friendly, use a clear visual hierarchy, and ensure the profile and feedback pages are easy to scan.

We will also consider emotional accessibility. Because simulation practice can feel stressful, especially for novice teachers, the platform should not make users feel judged or ranked. Larson et al. (2019) found that simulated teaching practice can be useful, but some participants also experienced stress and performance anxiety. Therefore, our feedback should be framed as growth-oriented rather than evaluative. Instead of saying that a teacher “failed,” we can explain what the choice supported, what it missed, and what the teacher might try next.

Finally, we will gather feedback through short user testing with professors, TFs, and classmates. We want to observe whether users understand the personality test, trust the identity result, know how to enter the simulation, and find the profile useful after completing a scenario. We will use their feedback to revise the scenario wording, the result page tone, the visual design, and the overall flow.

Conclusion

Teacher Reflection Lab 2.0 turns our second mini-project into a more complete online practice system for teacher growth. The final version will connect teacher personality, scenario-based decision-making, feedback, and profile-based reflection into one coherent learning journey. Our goal is not to tell teachers who they are, but to help them practice who they are becoming. By combining low-risk simulations with reflective teacher profiles, the platform supports novice and pre-service teachers in developing self-awareness, classroom judgment, and confidence before they face difficult moments in real classrooms.

References

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